

CS3240

Data Analysis

From Persona/Journey → Tasks → Themes → Quant Analysis

S2AY2526

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Lecture 4(L4)

Persona → WHO we design for

Journey → WHERE breakdowns happen

Lecture 7(L7)

Task Analysis → HOW the user actually performs the task

Thematic Analysis → WHAT patterns explain behavior

Quantitative Analysis → HOW MUCH / HOW OFTEN

Task

A task is a set of actions performed by a user to achieve a specific goal.

Goals → Tasks → Subtasks

Example:

Persona1: **Between-Classes Sprinter**
Context

- ≤ 2 minutes available
- Walking between classes
- Divided attention
- One-handed use

Goal: Get lunch quickly

Key Task: **Order food fast**

Subtasks:

- Open app
- Select usual order
- Confirm pickup
- Pay
- Leave immediately

Success Criteria:

- Completed in under 60–90 seconds
- Minimal thinking
- No unexpected interruptions

Example:

Persona 2: Planner & Explorer

Context

- Sitting down during free time
- Full attention
- No time pressure
- Often browsing on Wi-Fi

Goal :Make an informed and satisfying choice.

Key Task :**Compare food options and plan meal.**

Subtasks

- Open app
- Browse categories
- Filter by cuisine / price / wait time
- Compare 2–3 stalls
- View reviews / promotions
- Save favourites or decide
- Order later (optional)

Success Criteria

- Feels confident in decision
- Discovers good deals
- Can return later without losing options

How the Two Tasks Differ

	Sprinter	Planner
Time Pressure	High	Low
Primary Need	Speed	Information
Tolerance for Steps	Very low	Moderate
Browsing	Liability	Value
Navigation Style	Linear	Non-linear
Default vs Customisation	Defaults	Custom filters
Cognitive Load	Must be minimal	Can be higher
Task Length	3–4 steps	6–8 steps

What This Changes in the UI

For Sprinter

- “Order Usual” or “Order Favorite” button
- Recent orders surfaced
- Auto-filled payment
- No pop-ups mid-flow

For Planner

- Search bar (prominent)
- Filters & sorting
- “Compare” view
- Save / wishlist
- Promotions are visible

What is meant by 'Analysing a Task' (Task Analysis) ?

Visualize the task

- Break down each step into a clear representation (typically a digram)

Understand user perception

- What actions will users take first?
- What will they notice next?
- How will they move step by step through the process?

Evaluate current behavior

- Study how people already perform the task.
- Align design with their mental models & habits.

Identify pitfalls

- Spot frustrations and errors from current experiences.
- Ensure your design avoids repeating negative patterns.

Example - Task Analysis

Goal: Successfully place an order before next class.

Task: Order Food for Quick Pickup

Step Breakdown:

1. Open app
2. Select stall
3. Select item
4. Add to cart
5. Confirm pickup time
6. Pay

Clear start → clear end.

Identify Decision Points

Decision Point 1: Stall Available?

- Yes → Continue
- No → Choose alternative stall

Tasks are rarely straight lines.

Decision points shape design requirements.

Decision Point 2: Payment Successful?

- Yes → Order confirmed
- No → Retry or change method

Identify Pitfalls & Breakdowns

Where can this fail?

- ⚠ Hidden fees appear late
- ⚠ Long queue increases pickup time
- ⚠ Payment failure
- ⚠ Stall closes unexpectedly

Task analysis reveals
friction before UI design.

From Task Analysis → Design Decisions

Without Task Analysis:

- You design attractive screens.

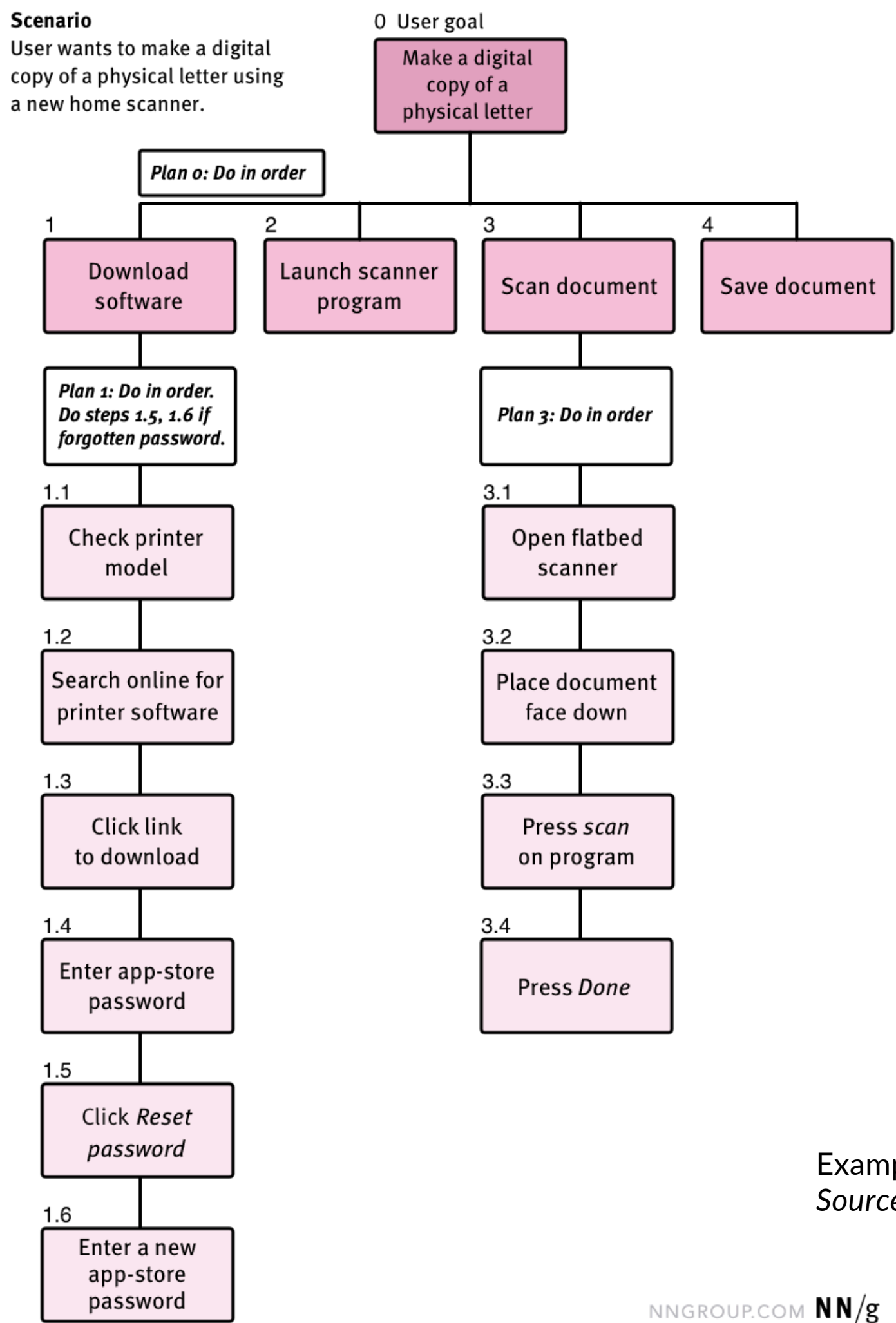
With Task Analysis:

- Show stall open/closed & queue status upfront
 - Show full price before checkout
 - Provide fast payment retry
 - Display pickup time clearly
- and more..

Design decisions become defensible.

Example

You can label sub-tasks for referencing during design process



Example:
Source: Task Analysis article at nnGroup.com

Why Tasks and Task Analysis Matter in UI/UX Design

User interaction with system = Task performance

A task:

- Is an observable activity
- Has a clear start and end
- Can succeed or fail
- **Design succeeds when tasks succeed.**

A persona WITHOUT task analysis = a story

A task analysis WITHOUT persona = mechanics

Together → defensible design decisions.

Example : Persona WITHOUT Task Analysis = Story

Persona Context :

Alicia is a busy student who values efficiency and hates wasting time. She prefers seamless digital experiences. She wants to use the app for ordering food.

Without task analysis, the team says:

- “Let’s make it clean.”
- “Let’s reduce clutter.”
- “Let’s make it modern.”

These are aesthetic decisions, not task decisions.

There is no:

- Step breakdown
- Decision prioritisation
- Flow optimisation

It’s empathy, not execution.

That’s a story.

Example: Task Analysis WITHOUT Persona = Mechanics

Task: Order food

Steps:

- Open app
- Browse
- Select item
- Add to cart
- Checkout
- Pay

Perfectly logical.

But for whom?

Is the user:

- Rushing?
- Exploring?
- Budget-sensitive?
- Comparing?

Without persona, we don't know:

- Whether browsing should exist
- Whether defaults matter
- Whether filters matter
- Whether comparison is necessary

It becomes a neutral machine workflow.
That's mechanics.

Example: Persona + Task Analysis = Defensible Decision

Persona: Between-Classes Sprinter

Context: ≤ 2 minutes

Goal: Complete fast

Task: Order Food

Task Analysis:

- Open app
- “Order Usual”
- Confirm
- Done

Browsing removed.

Now if a team member says:

Let's add a recommendation carousel before checkout.

It can be argued - a **defensible design decision**

-No. It interrupts the critical path for the Sprinter persona.

Identifying Key Tasks (Example)

Between-Classes Sprinter

Primary Context: ≤ 2 minutes, rushing

Key Tasks:

- Order food
- Pay quickly
- Check transport timing

Priority: Speed over exploration

Planner & Explorer

Primary Context: Downtime, full attention

Key Tasks:

- Compare options
- Save favourites
- Track promotions

Priority: Information & control

What Makes a Key Task Strong? (CS3240 project relevant)

Key Tasks Must:

- Be multi-step (not one-click actions)
- Be observable (clear start and end)
- Be grounded in user research evidence
- Influence UI structure and feature prioritisation

If the task does not change user Interactions or UI decisions, it is not a key task.

See Appendix for some examples from Student projects

Next let's look at Thematic Analysis

Raw notes → Clusters → Themes → Design Ideas

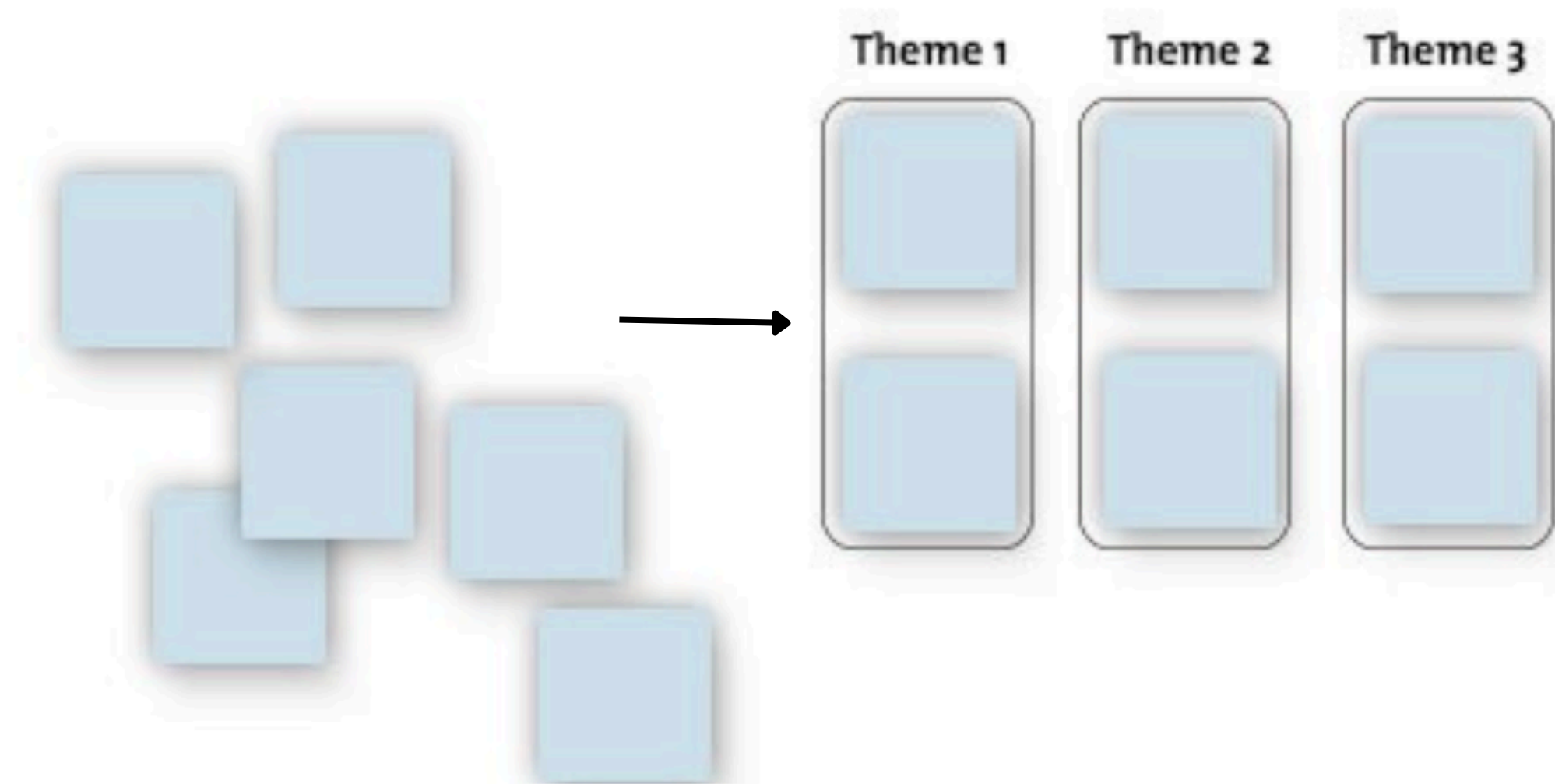
Themes are discovered from Patterns

Personas and Tasks also come from patterns

*We look for **patterns** across user data*

Thematic Analysis

A **clustering** exercise (**Affinity Diagramming**) in which you bundle and group information.



Example

U01-1

I have few close friends on Facebook. That's all I really care about.

U02-2

I only use Facebook to keep up with my mom.

U03-1

I wish I could completely hide my profile so no one else can find me.

U04-1

I am not available to make new friends.

user notes from 4 different users

Cluster(create **affinities** of) user notes based on common pattern/theme

U02-2 I only use Facebook to keep up with my mom.

U04-1 I am not available to make new friends.

U01-1 I have few friends on Facebook. That's all I really care about.

U03-1 I wish I could completely hide my profile so no one else can find me.

Next...

Attach a summarised Label for each cluster

Label

*I use facebook to keep
in touch with few
people close to me.*

U02-2 I only use
Facebook to keep
up with my mom.

U01-1 I have few
friends on
Facebook. That's
all I really care
about.

*I don't want to be
bothered by people
who are not close to
me.*

Label

U04-1 I am not
available to make new
friends.

U03-1 I wish I could
completely hide my
profile so no one else
can find me.

Next...

Continue to build bottom up hierarchy
-- typically 3 levels of labels

Some people want to share their social life with only few select and not with the whole world.

Label

I use facebook to keep in touch with few people close to me.

I don't want to be bothered by people who are not close to me.

U02-2 I only use Facebook to keep up with my mom.

U04-1 I am not available to make new friends.

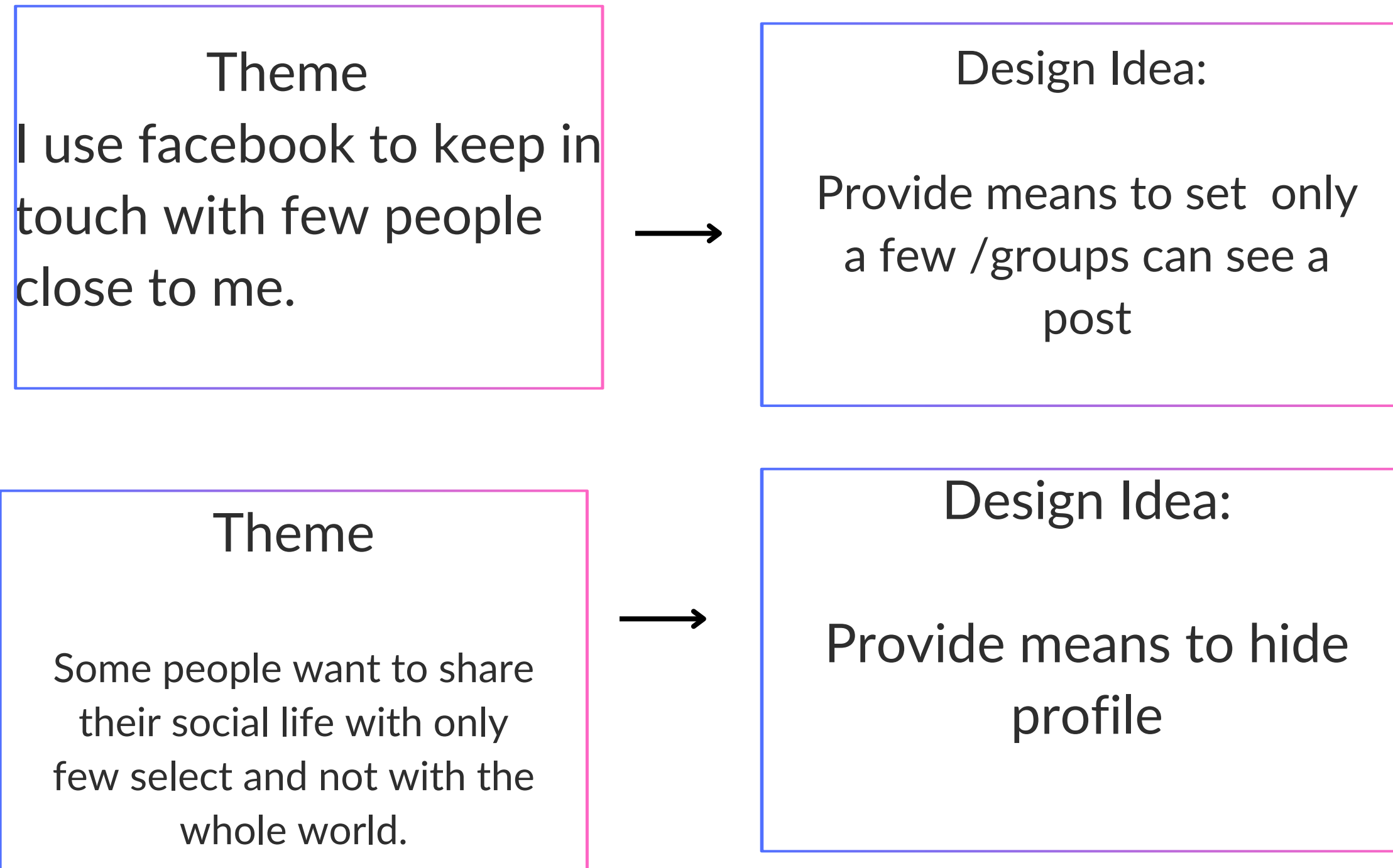
U01-1 I have few friends on Facebook. That's all I really care about.

U03-1 I wish I could completely hide my profile so no one else can find me.

What next ?

Each identified theme $\leftrightarrow$ a user **need**.

Each identified theme $\longrightarrow$ Design Idea (DI)



Some people want to share their social life with only few select and not with the whole world.

I use facebook to keep in touch with few people close to me.

U02-2 I only use Facebook to keep up with my mom.

U01-1 I have few friends on Facebook. That's all I really care about.

I don't want to be bothered by people who are not close to me.

U04-1 I am not available to make new friends.

U03-1 I wish I could completely hide my profile so no one else can find me.

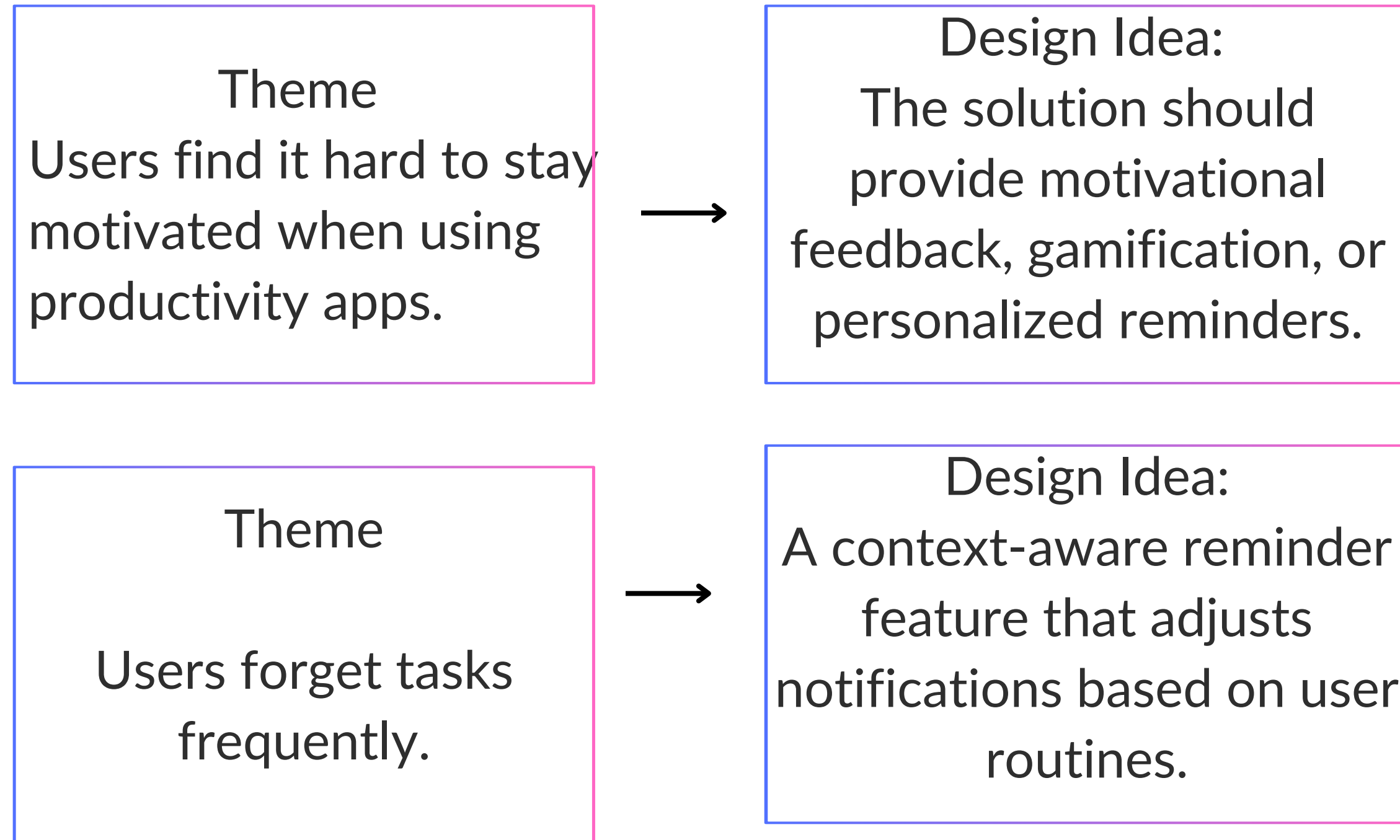
Design Idea (DI)

1 : Provide means to set only a few/ groups can see a post

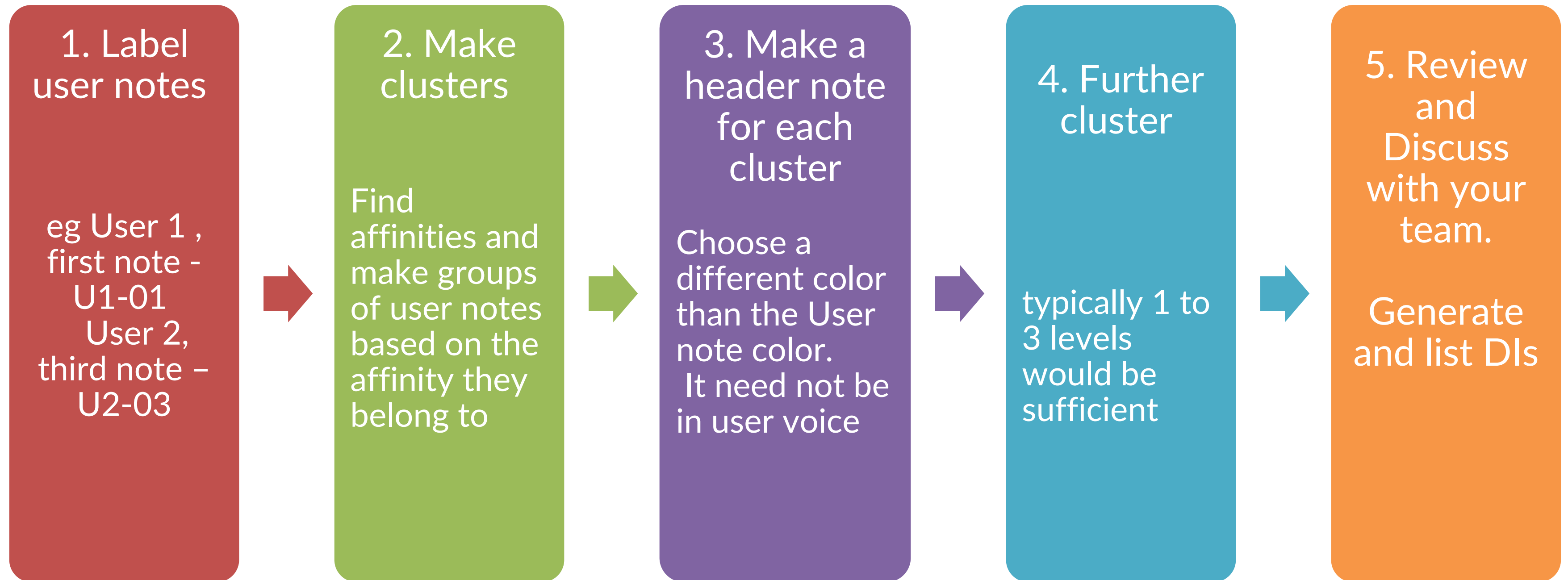
2 : Provide ways to hide a post

3 : Provide means to hide profile

Another Example of Theme to Design Idea



So far: Steps of the Affinity Diagram Process

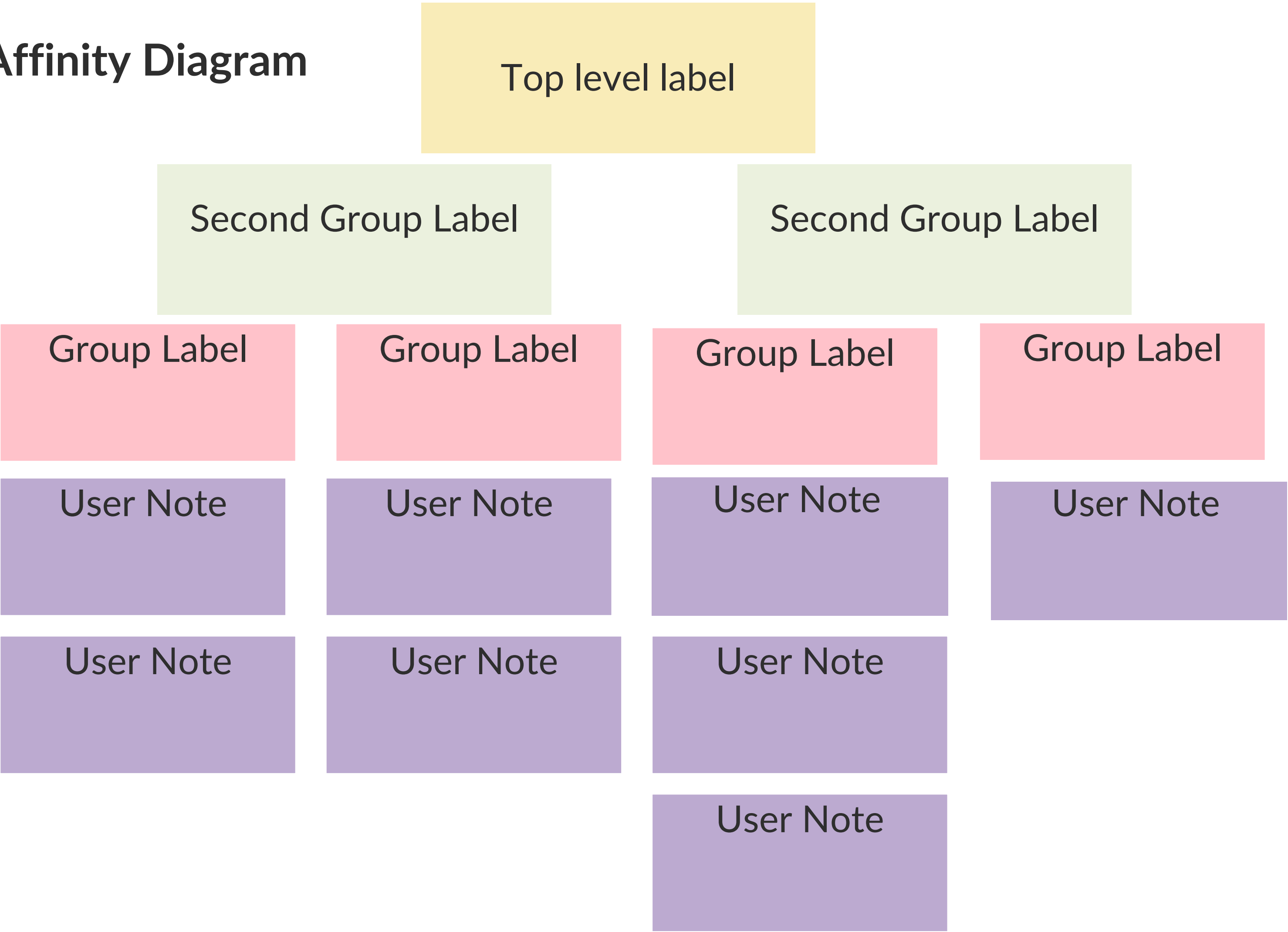


Reminders:

Cluster user notes to make '**affinities**' - e.g. cluster notes that focus on similar intents, problems and issues.

Affinity diagrams turn messy qualitative data into structured insights that directly guide design decisions.

A template for Affinity Diagram



See Appenidx for Student example and some references for Affinity diagram

AI in Identifying User tasks and Thematic Analysis

Identifying Key
User Tasks

Extract User
Notes

Cluster and
Label User Notes

Turn Themes
into Design Ideas

Create Affinity
Diagram

When we use AI for data analysis, we have to **keep it tied to the raw data**.

Do not ask it to “make up” findings or let it fill in gaps.

AI reasoning is based on patterns learned from its training data, which may not match your specific context or the reality of your study.

Use it to speed things up, but always double-check: go back to the quotes, notes, or numbers, and consider the real context.

Identifying Key User Tasks

AI needs raw user research data:

- User interview transcripts
- Survey answers
- Observation notes
- User goals or scenarios
- Pain points or frustrations
- User context

Prompt Template:

ROLE: Act as a senior UX researcher.

DATA: <interview quotes / notes / observation logs>

CONCEPT: <1 sentence product concept>

USER & GOAL: <target user and research goal>

TASK: Identify the key user tasks and sub-tasks based on the data.

OUTPUT: Bullet list with headings, including assumptions and supporting evidence.

Example:

Ai Dj For ...ew Results

 DOCX

ROLE: Act as a senior UX researcher.

DATA: Use this interview document.

CONCEPT: Spotifywants to add an"AI DJ for Study" mode that learns astudent's study patterns and automatically buildsplaylists (and gentle transitions) to help them focus, without needing them to manually search or skip tracks.

USER & GOAL: Understand how universitystudents currently choose music while studying, what"focus" means to them, and what would make themtrust (or reject) an AI-driven study DJ.

TASK: Identify the key user tasks and sub-tasks based on the data.

OUTPUT: Bullet list with headings, including assumptions and supporting evidence.

We can upload the user research notes document and ask AI to use it directly.

Gemini 3

1. Establishing the Initial Study Environment

Users invest significant cognitive effort into "setting the stage" before they begin their actual academic work.

- **Sub-task: Curating music based on task type or mood**
 - **Evidence:** Daniel selects specific fixed playlists based on whether he is coding or reading. Aisha manually switches playlists to align with her creative phase. [🔗](#) [🔗](#)
 - **Assumption:** Users believe that the "wrong" genre of music will actively hinder their ability to start a task.
- **Sub-task: Searching for the "perfect" track to initiate focus**
 - **Evidence:** Mei spends considerable time searching for the right playlist before she can actually start studying. [🔗](#)
 - **Assumption:** Reducing "search time" is a high-value opportunity for the AI DJ to minimize pre-study procrastination.

Maintaining "Flow" and Minimizing Distraction

ce a study session begins, the primary task shifts from selection to maintenance. The goal is keep music in the "perceptual background."

Sub-task: Eliminating manual interface interactions

- **Evidence:** Mei considers "focus" to mean not checking her phone. Ryan defines focus breakers as any moment he is forced to interact with the app to skip or check songs. [🔗](#) [🔗](#) [🔗](#)
- **Assumption:** Every time a user touches their phone to adjust music, there is a high risk of "context switching" into social media or other distractions.

Sub-task: Managing auditory transitions

- **Evidence:** Aisha is sensitive to harsh or abrupt transitions between moods. Mei is distracted by sudden emotional or dramatic song changes. [🔗](#) [🔗](#) [🔗](#) [🔗](#) [🔗](#)
- **Assumption:** The way songs connect is more critical to maintaining flow than the specific songs themselves. [🔗](#)

3. Calibrating AI Automation and Trust

Users need to feel they are collaborating with the AI rather than being controlled by it, especially during high-stakes tasks.

- **Sub-task: Verifying AI logic (Transparency)**
 - **Evidence:** Daniel requires clear reasons for why the AI is changing the music to trust the system. [🔗](#) [🔗](#) [🔗](#)
 - **Assumption:** A "black box" approach to music selection creates anxiety for productivity-oriented users.
- **Sub-task: Overriding or "Nudging" the AI**
 - **Evidence:** Mei would reject the system if it felt difficult to turn off. Users generally want automation but demand control during critical moments. [🔗](#) [🔗](#) [🔗](#) [🔗](#)
 - **Assumption:** Users need a "panic button" or a simple way to revert to manual mode without breaking their concentration.

The framing could be more user-centered.

Some terms feel abstract, so simplifying the wording would make it more accessible.

4. Defining and Monitoring the "Focus State"

The user's task is not just to study, but to monitor their own mental state to ensure they are being efficient.

- **Sub-task: Assessing cognitive load vs. music intensity**
 - **Evidence:** Daniel is concerned about the AI misjudging the intensity of his engineering tasks. Mei finds that lyrics in familiar languages break her concentration. [🔗](#) [🔗](#) [🔗](#) [🔗](#)
 - **Assumption:** The AI must distinguish between "passive background noise" (Ryan) and "active emotional alignment" (Aisha). [🔗](#) [🔗](#)

Would you like me to translate these tasks into a set of User Stories or a preliminary Feature Roadmap for the Spotify development team?

Using the following prompt makes the results clearer and easier to understand, as it instructs the AI to avoid overly abstract and technical terminology.

Avoid abstract or academic language.

Describe what the user actually does, step by step, using simple verbs.

Imagine explaining the behavior to a friend who has never studied UX.

Each sub-task should describe a real action or decision the user makes.



To help our team build this, I've broken down exactly what these students do when they sit down to work. Think of this as a "play-by-play" of their study habits and how they decide whether to let the AI take the wheel. [🔗](#) [🔗](#)

1. Getting Ready to Work

Before they even open a book, students spend time "setting the stage" with their music. [🔗](#)



- **Hunting for a vibe:** They scroll through lo-fi or study playlists to find one that feels right. [🔗](#)
- **Matching the music to the job:** They pick specific songs based on what they are doing, like choosing one playlist for coding and a different one for reading. [🔗](#) [🔗](#)
- **Checking their mood:** They switch up the music to match how they feel emotionally before they start. [🔗](#)
- **Assumptions:** If they can't find the "perfect" song quickly, they might keep searching instead of actually studying. [🔗](#)

2. Trying to Stay "In the Zone"

Once they start working, their main goal is to forget the music is even playing. [🔗](#) [🔗](#) [🔗](#)

- **Ignoring the phone:** They try to stop checking their phone or looking at the app. [🔗](#) [🔗](#)
- **Blocking out words:** They actively avoid songs with lyrics they recognize so they don't start singing along. [🔗](#) [🔗](#) [🔗](#) [🔗](#)
- **Avoiding the "skip" button:** They try to pick music that won't make them want to reach over and skip a track. [🔗](#) [🔗](#)
- **Assumptions:** Every time a student has to touch their phone to fix a song, they get distracted and lose their momentum. [🔗](#) [🔗](#)

3. Dealing with Interruptions

When the music changes or gets annoying, the student has to make a split-second choice. 



- **Reacting to a "bad" song:** If a song is too loud, too fast, or too emotional, they have to stop working to change it.   
- **Checking the AI's "brain":** If the AI changes the song, they look at the screen to see *why* it made that choice.   
- **Taking back control:** If they don't like what the AI is doing, they turn it off or go back to their old playlists.   
- **Assumptions:** If the AI makes even one or two "weird" choices in a row, the student will stop trusting it and go back to doing it manually.   

4. Judging if the AI is Helpful

In the end, the student decides if the AI DJ is actually making their life easier.  

- **Counting the clicks:** They notice if they are spending less time fiddling with the app.    
- **Noticing the "flow":** They feel out whether the music transitions are smooth or if they are "jarring" and annoying.   
- **Assumptions:** If the AI is too "needy" (asking too many questions or requiring setup), the student will find it more annoying than helpful. 

Would you like me to create a "Day in the Life" story for one of these students to show exactly where the AI DJ would help them most?

Extract User Notes

Prompt for extracting user notes from interview data:

For Affinity Diagram the first step is to have the AI generate different users' viewpoints in a fixed, consistent format.

Ai Dj For ...ew Results

W DOCX

TASK:

Based on the interview document, extract and rewrite the interview data into short, atomic user notes, suitable for placing on sticky notes.

RULES:

Each note must represent a clear idea, feeling, behavior, or need.

Use first-person voice.

Keep each note to 1–2 short sentences.

Do NOT summarize across users. Do NOT merge ideas.

Stay close to what the user actually said; avoid researcher language.

LABELING FORMAT:

Label each note as U{user number}-{note number}

Example: U02-3 = third note from User 2

Number notes sequentially per user.



Here are the atomic user notes extracted from the interview data, formatted as individual sticky notes for your research synthesis.

P1 — Mei (Social Sciences)

- **U1-1:** I often spend about 10 minutes just searching for the "right" playlist before I can actually start my work.
 - **U1-2:** Even when I use study-specific playlists, I still find myself skipping a lot of tracks.
 - **U1-3:** I get distracted if a song has lyrics in a language I understand too well.
 - **U1-4:** To me, being focused means I don't notice time passing and I'm not checking my phone constantly.
 - **U1-5:** When I am in the zone, I want the music to just disappear into the background.
 - **U1-6:** Sudden changes in mood—like a chill track turning dramatic—really pull me out of my focus.
 - **U1-7:** If a song comes on that I really like, I start paying attention to it instead of my study material.
 - **U1-8:** I like the idea of an AI DJ if it means I don't have to keep touching my phone.
 - **U1-9:** I will trust the system if it learns what specifically distracts me rather than just what I like.
 - **U1-10:** I would reject this if it feels random or if I can't easily turn it off.
-

P2 — Daniel (Engineering)

- **U2-1:** I have two specific, fixed playlists for coding and reading; I don't really explore new music while working.
 - **U2-2:** Focus for me is all about efficiency—finishing problem sets faster with fewer mistakes.
 - **U2-3:** Lyrics, tempo changes, or unpredictable recommendations in the middle of a session break my concentration.
 - **U2-4:** I'm cautious because I don't trust an AI to know when I need intense focus versus light background sound.
 - **U2-5:** I would trust the AI more if it explained the logic behind its choices, like telling me how long I've been studying.
 - **U2-6:** I will reject the tool if it keeps experimenting while I am in the middle of something important.
-

P3 — Aisha (Design)

- **U3-1:** I manually change my playlists frequently based on my current mood or the specific design phase I'm in. ⓘ
 - **U3-2:** Being focused means being "in the zone" and emotionally aligned with the work I'm creating. ⓘ
 - **U3-3:** Harsh transitions where one song feels like a "different world" from the last one break my flow. ⓘ
 - **U3-4:** I'm interested in an AI that feels more like a real DJ with gentle transitions between tracks. ⓘ
 - **U3-5:** I will trust this if it adapts slowly and respects my current mood. ⓘ
 - **U3-6:** I'll reject it if it pushes a "productivity" logic that ignores my creative rhythm. ⓘ
-

Based on these results, the clustering is clear, which indicates that our prompt is effective.

P4 — Ryan (Business)

- **U4-1:** I just hit play on whatever I last listened to because music is mostly just background noise for me. ⓘ
 - **U4-2:** Focus just means that I'm staying awake and I'm not bored. ⓘ
 - **U4-3:** My focus breaks the moment I have to start interacting with the app to skip songs or check what's playing. ⓘ
 - **U4-4:** If an AI DJ means I don't have to touch or think about my phone while studying, I consider that a win. ⓘ
 - **U4-5:** I'll trust it as long as it works quietly in the background without needing much from me. ⓘ
 - **U4-6:** I will reject it if it asks me too many questions or requires a new setup every single time I use it. ⓘ
-

Would you like me to organize these notes into a table to compare how different students define "focus"?

Cluster and Label User Notes

Prompt for clustering and labelling:

Group these notes into clusters. For each cluster:

- 1) Give a short, clear label written as a first-person insight sentence.
- 2) List the user quotes that belong to that cluster.



I have grouped the sticky notes into five main clusters based on the students' shared behaviors and needs. Each group represents a core insight into how they experience study music.

"I want my music to be a background layer that doesn't demand my attention."

- **U1-5:** When I am in the zone, I want the music to just disappear into the background. 
- **U1-7:** If a song comes on that I really like, I start paying attention to it instead of my study material. 
- **U4-1:** I just hit play on whatever I last listened to because music is mostly just background noise for me. 
- **U4-5:** I'll trust it as long as it works quietly in the background without needing much from me. 

"Any time I have to touch the app or my phone, my focus is ruined."

- **U1-4:** To me, being focused means I don't notice time passing and I'm not checking my phone constantly. [↗](#)
- **U1-8:** I like the idea of an AI DJ if it means I don't have to keep touching my phone. [↗](#)
- **U4-3:** My focus breaks the moment I have to start interacting with the app to skip songs or check what's playing. [↗](#)
- **U4-4:** If an AI DJ means I don't have to touch or think about my phone while studying, I consider that a win. [↗](#)
- **U4-6:** I will reject it if it asks me too many questions or requires a new setup every single time I use it. [↗](#)

"Sudden or unpredictable changes in the sound pull me out of my work."

- **U1-6:** Sudden changes in mood—like a chill track turning dramatic—really pull me out of my focus. [↗](#)
- **U2-3:** Lyrics, tempo changes, or unpredictable recommendations in the middle of a session break my concentration. [↗](#)
- **U3-3:** Harsh transitions where one song feels like a "different world" from the last one break my flow. [↗](#)
- **U3-4:** I'm interested in an AI that feels more like a real DJ with gentle transitions between tracks. [↗](#)

"I need the music to match the specific difficulty or mood of my task."

- **U2-1:** I have two specific, fixed playlists for coding and reading; I don't really explore new music while working. [↗](#)
- **U2-4:** I'm cautious because I don't trust an AI to know when I need intense focus versus light background sound. [↗](#)
- **U3-1:** I manually change my playlists frequently based on my current mood or the specific design phase I'm in. [↗](#)
- **U3-2:** Being focused means being "in the zone" and emotionally aligned with the work I'm creating. [↗](#)
- **U3-6:** I'll reject it if it pushes a "productivity" logic that ignores my creative rhythm. [↗](#)

"I struggle with the 'setup' time required to start a productive session."

- **U1-1:** I often spend about 10 minutes just searching for the "right" playlist before I can actually start my work. [↗](#)
- **U1-2:** Even when I use study-specific playlists, I still find myself skipping a lot of tracks. [↗](#)
- **U1-9:** I will trust the system if it learns what specifically distracts me rather than just what I like. [↗](#)
- **U2-5:** I would trust the AI more if it explained the logic behind its choices, like telling me how long I've been studying. [↗](#)

Would you like me to identify the "How Might We" statements for these clusters to help jumpstart the design phase?

Continue to build bottom up hierarchy:

Continue building a bottom-up affinity hierarchy.
Group these existing clusters into higher-level themes based on shared underlying needs or goals.

Different user viewpoints belong to different categories. When we want to progressively synthesize them into shared characteristics, it is necessary to give the AI clear guidance and explicit requirements.



To move from your observations to a higher-level strategy, I've grouped the existing clusters into **three core themes**. These represent the big-picture goals that will determine if these students stick with the AI DJ or delete it.

Theme 1: The Goal of "Invisible Automation"

This theme combines the need for a background presence with the desire to stop fiddling with technology. The underlying goal is to reduce the "mental tax" of managing music so the user can devote 100% of their brainpower to their task. [🔗](#) [🔗](#)

- **Cluster: I want my music to be a background layer that doesn't demand my attention.**
 - "The music kind of disappears into the background." [🔗](#)
 - "If Spotify throws in a song I really like—I start paying attention to it instead of my work." [🔗](#)
 - "Music is mostly background noise for me." [🔗](#)
 - "I'd trust it if it just works quietly in the background." [🔗](#)
- **Cluster: Any time I have to touch the app or my phone, my focus is ruined.**
 - "When I'm focused... I'm not checking my phone every few minutes." [🔗](#)
 - "Honestly, that sounds really nice. If it means I don't have to keep touching my phone." [🔗](#)
 - "When I start interacting with the app—skipping songs, checking what's playing... [that breaks my focus]." [🔗](#)
 - "If it means I don't have to touch my phone, that's a win." [🔗](#)
 - "I'd reject it if it asks me too many questions or needs setup every time." [🔗](#)

Theme 2: Predictability and Flow Protection

This theme brings together the fear of "surprises" and the need for smooth transitions. The underlying goal is **safety**; users want to know the AI won't "betray" them by injecting a distraction when they are most vulnerable. [🔗](#) [🔗](#) [🔗](#)

- **Cluster: Sudden or unpredictable changes in the sound pull me out of my work.**
 - "Sudden mood changes. Like when a chill track suddenly turns dramatic or emotional." [🔗](#)
 - "Anything unpredictable. Lyrics, tempo changes, or when Spotify 'recommends' something new in the middle." [🔗](#)
 - "Harsh transitions. If one song ends and the next feels like a different world, it breaks my flow." [🔗](#)
 - "I like the idea of gentle transitions, because sudden changes really pull me out." [🔗](#)
 - **Cluster: I struggle with the 'setup' time required to start a productive session.**
 - "Sometimes I spend like 10 minutes just finding the 'right' playlist before I even start studying." [🔗](#)
 - "I still end up skipping a lot [of tracks]." [🔗](#)
 - "I'd trust it if it learns what distracts me, not just what I like." [🔗](#)
 - "I'd trust it more if it explains why it's changing music." [🔗](#)
-

Theme 3: Dynamic Context Matching

This theme is about the AI's ability to understand *what* the student is doing. The underlying goal is **alignment**—ensuring the "vibe" of the music matches the cognitive load or creative requirements of the specific task. [🔗](#) [🔗](#)

- **Cluster: I need the music to match the specific difficulty or mood of my task.**
 - "I have two playlists. One for coding, one for reading." [🔗](#)
 - "I don't fully trust AI to know when I need intense focus versus lighter background sound." [🔗](#)
 - "Sketching is different from polishing a final design. I manually change playlists a lot based on my mood." [🔗](#)
 - "Focus... [is] being 'in the zone.' Not silent, but emotionally aligned with what I'm making." [🔗](#)
 - "I'd reject it if it pushes productivity too hard and ignores creative rhythm." [🔗](#)

Would you like me to turn these high-level themes into a set of "User Requirements" for the product's MVP (Minimum Viable Product)?

Themes to Design Ideas

Prompt for generating design ideas:

For each identified theme mentioned above, generate 1–2 Design Ideas.

Each Design Idea should:

Directly respond to the user's need or problem expressed in the theme

Be actionable and designable (not abstract principles)

Describe what the product could do, not how it is technically implemented

Here are specific design ideas for the Spotify "AI DJ for Study" based on the high-level themes identified from the student interviews.

Theme 1: The Goal of "Invisible Automation"

The Problem: Students lose focus the moment they have to interact with their phone or the app to fix the music. [🔗](#) [🔗](#) [🔗](#) [🔗](#) [🔗](#)

- **Design Idea: "One-Tap Study Lock"** Upon starting a session, the app enters a simplified, distraction-free interface that hides social features and track lists. It replaces the skip button with a "Not my vibe" button that instantly adjusts the current sound profile without requiring the user to scroll through alternatives or see their notification tray. [🔗](#) [🔗](#) [🔗](#)
[🔗](#) [🔗](#)
- **Design Idea: "Hands-Free Session Extension"** If the AI detects that the user is still in a high-focus state (e.g., no phone movement or app interaction), it automatically extends the current music style past the original playlist end-point to avoid silence or a jarring shift to "Daily Mix" music. [🔗](#) [🔗](#) [🔗](#)

Theme 2: Predictability and Flow Protection

The Problem: Unexpected changes in tempo, lyrics, or mood break a student's "flow" and make them distrust the AI. [🔗](#) [🔗](#) [🔗](#) [🔗](#)

- **Design Idea: "Focus-Matched Crossfader"** The DJ uses extra-long, beat-matched transitions between tracks to ensure there is never a "silence gap" or a sudden shift in energy. The user can set a "Mood Ceiling" to prevent the AI from ever playing a track that exceeds a certain intensity or introduces vocals during a reading task. [🔗](#) [🔗](#) [🔗](#) [🔗](#)
[🔗](#)
- **Design Idea: "The Why-Bubble"** When the AI DJ makes a shift in the music, a small, non-intrusive text snippet appears on the lock screen (e.g., "*Slowing down for your deep work block*" or "*40 minutes in—gentle shift for a break*"). This gives the user the transparency they need to trust the system's logic. [🔗](#)

Theme 3: Dynamic Context Matching

The Problem: Students have different music needs depending on whether they are doing "intense" work like engineering or "creative" work like design.     

- **Design Idea: "Task-Intensity Slider"** Instead of choosing a genre, the user selects their task type (e.g., "Problem Sets," "Sketching," "Admin"). The AI DJ then adjusts the complexity of the music—playing minimalist, repetitive sounds for "Heavy Thinking" and more emotionally resonant, varied tracks for "Creative Flow".     
- **Design Idea: "Mood-First Onboarding"** When the study mode begins, the DJ plays 10 seconds of a "vibe" and asks for a simple thumbs up/down. This allows the student to align the AI with their current emotional state immediately, preventing the need for manual searching or skipping later in the session.     

Would you like me to draft a user testing plan to see which of these ideas resonates most with students like Mei and Daniel?

Create Affinity Diagram

Prompt for generating picture:

TASK:

Visualize the following affinity hierarchy as a clean, top-down diagram.

VISUAL RULES:

- Top-Level Theme at the top center
- Second-Level Themes beneath it
- Group Labels below each theme
- User Notes at the bottom
- No crossing lines
- Clear vertical hierarchy

Do not add or change any content.
Only focus on layout clarity and hierarchy.

We can first feed the AI with the previously generated insights from the Themes to Design Ideas step, and then add the above words, instructing it to generate corresponding visuals.

Remember this:

Here are specific design ideas for the Spotify "AI DJ for Study" based on the high-level themes identified from the student interviews.

Theme 1: The Goal of "Invisible Automation"

The Problem: Students lose focus the moment they have to interact with their phone or the app to fix the music.
Design Idea: "One-Tap Study Lock"Upon starting a session, the app enters a simplified, distraction-free interface that hides social features and track lists. It replaces the skip button with a "Not my vibe" button that instantly adjusts the current sound profile without requiring the user to scroll through alternatives or see their notification tray.

Design Idea: "Hands-Free Session Extension"If the AI detects that the user is still in a high-focus state (e.g., no phone movement or app interaction), it automatically extends the current music style past the original playlist end-point to avoid silence or a jarring shift to "Daily Mix" music.

Theme 2: Predictability and Flow Protection

The Problem: Unexpected changes in tempo, lyrics, or mood break a student's "flow" and make them distrust the AI.

Design Idea: "Focus-Matched Crossfader"The DJ uses extra-long, beat-matched transitions between tracks to ensure there is never a "silence gap" or a sudden shift in energy. The user can set a "Mood Ceiling" to prevent the AI from ever playing a track that exceeds a certain intensity or introduces vocals during a reading task.

Design Idea: "The Why-Bubble"When the AI DJ makes a shift in the music, a small, non-intrusive text snippet

shift in the music, a small, non-intrusive text snippet appears on the lock screen (e.g., "Slowing down for your deep work block" or "40 minutes in—gentle shift for a break"). This gives the user the transparency they need to trust the system's logic.

Theme 3: Dynamic Context Matching

The Problem: Students have different music needs depending on whether they are doing "intense" work like engineering or "creative" work like design.
Design Idea: "Task-Intensity Slider"Instead of choosing a genre, the user selects their task type (e.g., "Problem Sets," "Sketching," "Admin"). The AI DJ then adjusts the complexity of the music—playing minimalist, repetitive sounds for "Heavy Thinking" and more emotionally resonant, varied tracks for "Creative Flow".
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TASK:

Visualize the following affinity hierarchy as a clean, top-down diagram.

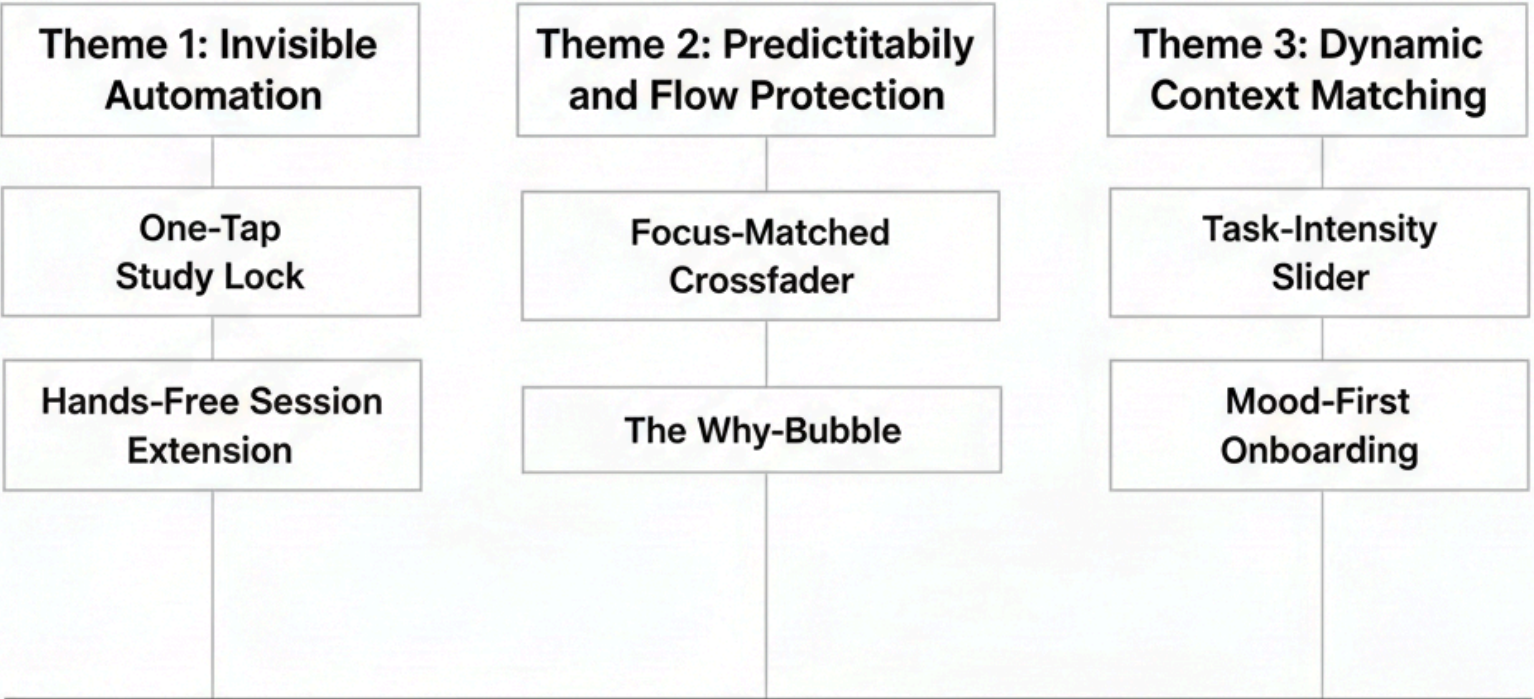
VISUAL RULES:

- Top-Level Theme at the top center
- Second-Level Themes beneath it
- Group Labels below each theme
- User Notes at the bottom
- No crossing lines
- Clear vertical hierarchy

Do not add or change any content.
Only focus on layout clarity and hierarchy.

The generated diagram lacks sufficient content richness to reflect the amount of upfront research work. The layout is basic, the text clarity is limited, and there are several spelling errors.

Spotify “AI DJ for Study” Insights



User Notes (Student Interview Takeaways)

- Students lose focus with phone interaction.
- Diffeexcted changes need different music.
- Need transparency to trust AI
- Want to avoid silence/jarring shifts

Another Prompt:

TASK:

Generate a hierarchical insight diagram for the concept "Spotify AI DJ for Study".

STRUCTURE REQUIREMENTS:

- Top level: High-level Themes
- Middle level: Design Ideas under each theme
- Bottom level: Detailed user notes (interview takeaways)

USER NOTES REQUIREMENTS:

- User notes must be detailed, specific, and written as concrete observations or quotes.
- Each user note must clearly support ONE design idea above it.
- Do NOT summarize user notes into generic bullets.
- Avoid vague statements such as "users want better focus".
- User notes should describe behaviors, frustrations, or expectations in context.

HIERARCHY RULES:

- Every Design Idea must have at least 2–3 supporting user notes.
- User notes must be visually and logically aligned with the Design Idea and Theme above.
- No orphan notes or floating insights.

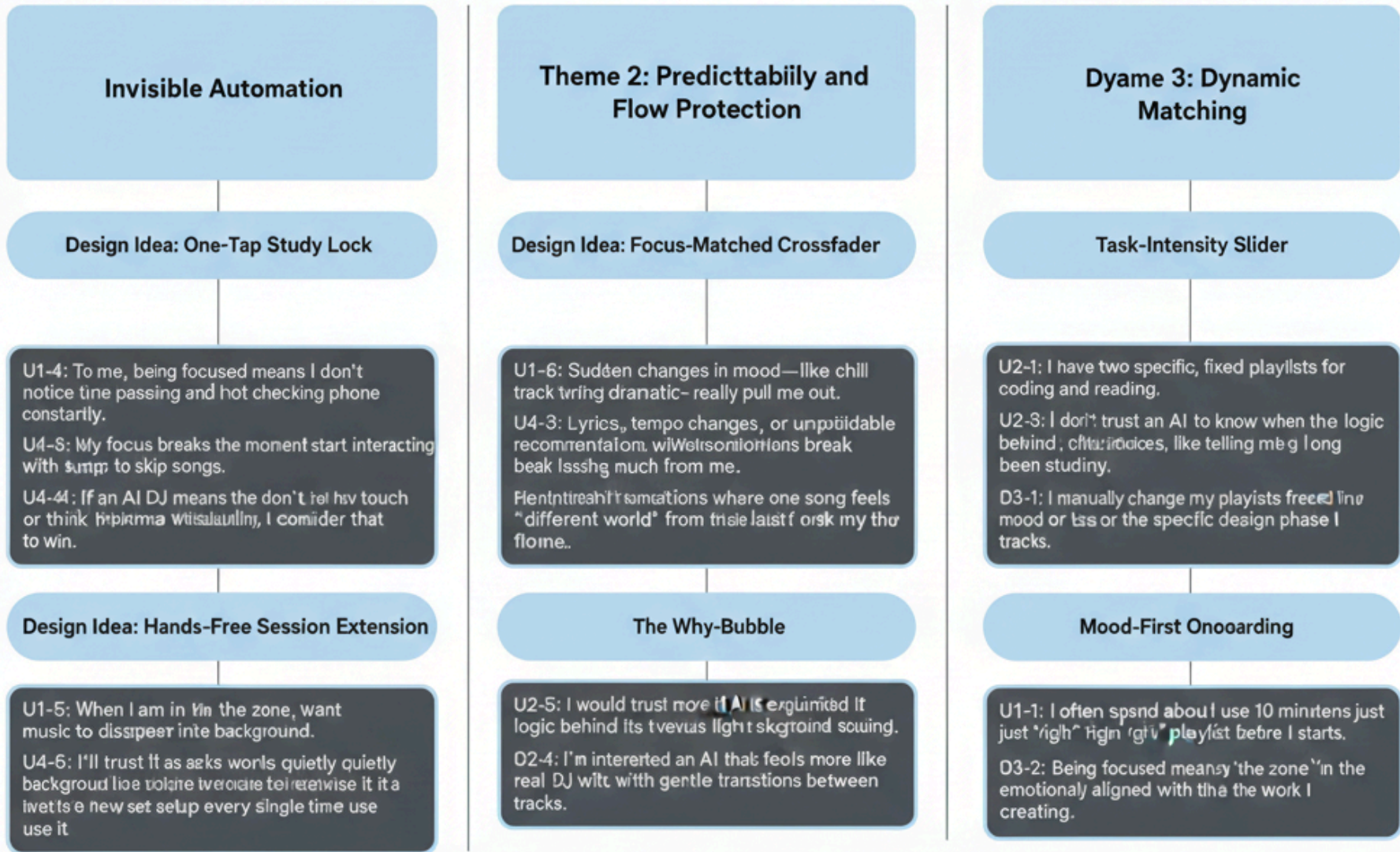
VISUAL OUTPUT REQUIREMENTS:

- Clearly show vertical alignment between Theme → Design Idea → User Notes
- Use indentation, grouping, or layout blocks to make relationships explicit
- Do not collapse user notes into a single shared list

OUTPUT:

A structured, diagram-like hierarchy where each user note explicitly justifies the design ideas above it.

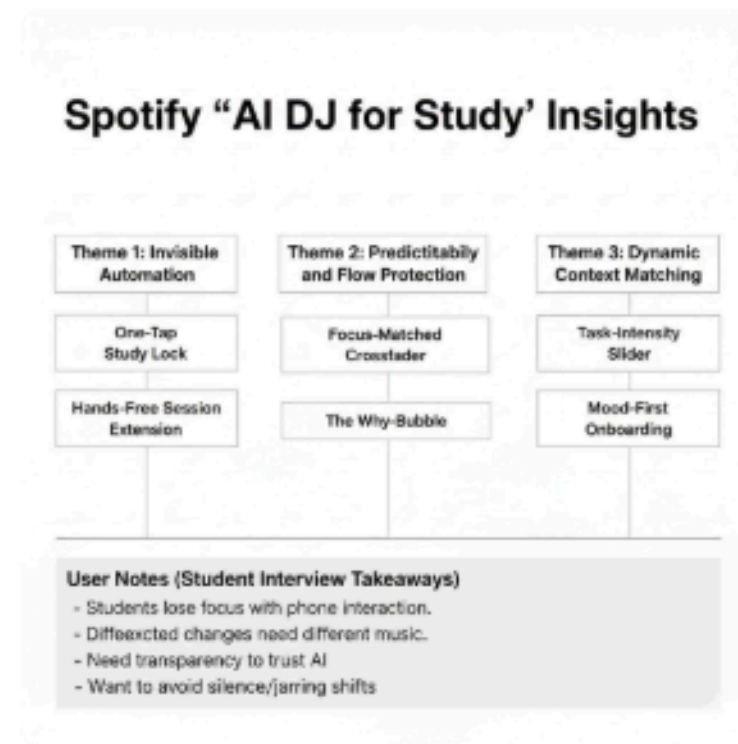
Spotify “AI DJ for Study” User Insights & Design Mapping



After switching to a more detailed prompt, we can observe that the final output is noticeably better than before. However, issues such as spelling errors and occasional ambiguity still remain.

Even after requesting revisions to address errors, the regenerated outputs still contain inaccuracies in content placement and logic. As a result, the visualization does not yet appear fully credible and should be treated as a reference only. The final diagram will therefore be manually created.

Tips:



This picture is too ordinary now. Can you write a better prompt?
The user note below needs to be very detailed and correspond one-to-one with the hierarchical structure above.

As a practical tip, when the output generated by Gemini is unsatisfactory and you are unsure how to adjust the prompt, you can ask another AI—such as ChatGPT—to review the result and suggest how the prompt should be revised to achieve the desired outcome.

Yes — and you’re asking for exactly the **right fix**.

The problem isn’t visual polish, it’s that the **evidence layer is too weak and generic**. What you need is a prompt that **forces traceability**: every user note must *justify* something above it.

Here’s a **much stronger prompt** you can use. It is written to **prevent generic bullets**, **enforce one-to-one mapping**, and **produce research-grade content**.

✓ Improved Prompt: Hierarchical Insight Diagram with Strong Evidence

sql

Copy code

ROLE: Act **as** a UX researcher **and** insight synthesizer.

TASK:

Generate a hierarchical insight diagram **for** “Spotify AI DJ **for** Study”.

The diagram must include three levels:

- 1) High-level Themes
- 2) Design Ideas under **each** theme
- 3) Detailed **User** Notes (interview takeaways) **at** the bottom

CRITICAL REQUIREMENTS **FOR USER NOTES**:

- **User** notes must be detailed, **specific**, **and** grounded **in** concrete behaviors **or** situations.
- **Each user** note must clearly **and** directly support **ONE** design idea above it.
- Do **NOT** write generic summaries (e.g., “users want better focus”).
- Write **user** notes **as** if they were distilled **from real** interview quotes **or** observations.
- **Each** design idea must have **2-3 unique** supporting **user** notes.
- **No user** note may support more than **one** design idea.

GPT 5.2

HIERARCHY RULES:

- Maintain strict **one-to-one** vertical alignment:
Theme → Design Idea → Supporting **User** Notes
- Avoid collapsing **all user** notes **into** a shared list.
- The hierarchy should read **as** a clear chain **of** reasoning **from** evidence **to** concept.

CONTENT GUIDELINES:

- Use plain, non-technical language.
- Avoid abstract **or** academic wording.
- Focus **on** actions, frustrations, expectations, **and** context.
- Treat **user** notes **as** evidence, **not** conclusions.

VISUAL / STRUCTURAL OUTPUT:

- Present the **result in** a clean, diagram-like structure **using** indentation **or** grouped blocks.
- Make the relationships **between** levels immediately readable.
- Do **not add** extra explanations outside the hierarchy.

OUTPUT FORMAT:

Theme:

Design Idea:

- **User** Note
- **User** Note
- **User** Note

🧠 Why this prompt works (and the previous one didn't)

- It **forbids generic notes** → no more "students lose focus"
- It **forces evidence ownership** → every note has a parent
- It **prevents floating insights** → nothing exists without justification
- It mirrors how **real affinity diagrams are assessed** in design education

✨ Optional add-on (if you want even higher quality)

Add this single line to the prompt:

pgsql

Copy code

- **If** a **user** note cannot clearly justify a design idea, **do not include** it.

This dramatically improves rigor.

One sentence you can use in your report

By requiring detailed, one-to-one mapped user notes, the revised prompt strengthens the evidential link between interview data, themes, and design ideas, making the hierarchy more credible and interpretable.

If you want, next I can:

- Rewrite the **existing diagram content** using this prompt
- Or help you turn one column into a **full concept slide** (research → insight → DI → value)

📄 👍 🗨️ ↺ ...

Quantitative Data Analysis

Numbers Tell a Story Too

Quantitative data complements qualitative findings, while qualitative gives you stories, quantitative gives you scale and evidence.

What can quantitative data tell us?

Quantitative data can tell us more about the usage analytics - **how many, how often, what and when** of e.g. a product's usage.

- How many users begin and finish a process?
- When is a certain option used?
- How often is an option used?
- Which different types of options do users select?
- What do users click on when they are on the main page?

Quantitative data cannot tell us the “why”

Quantitative data analysis

Averages:

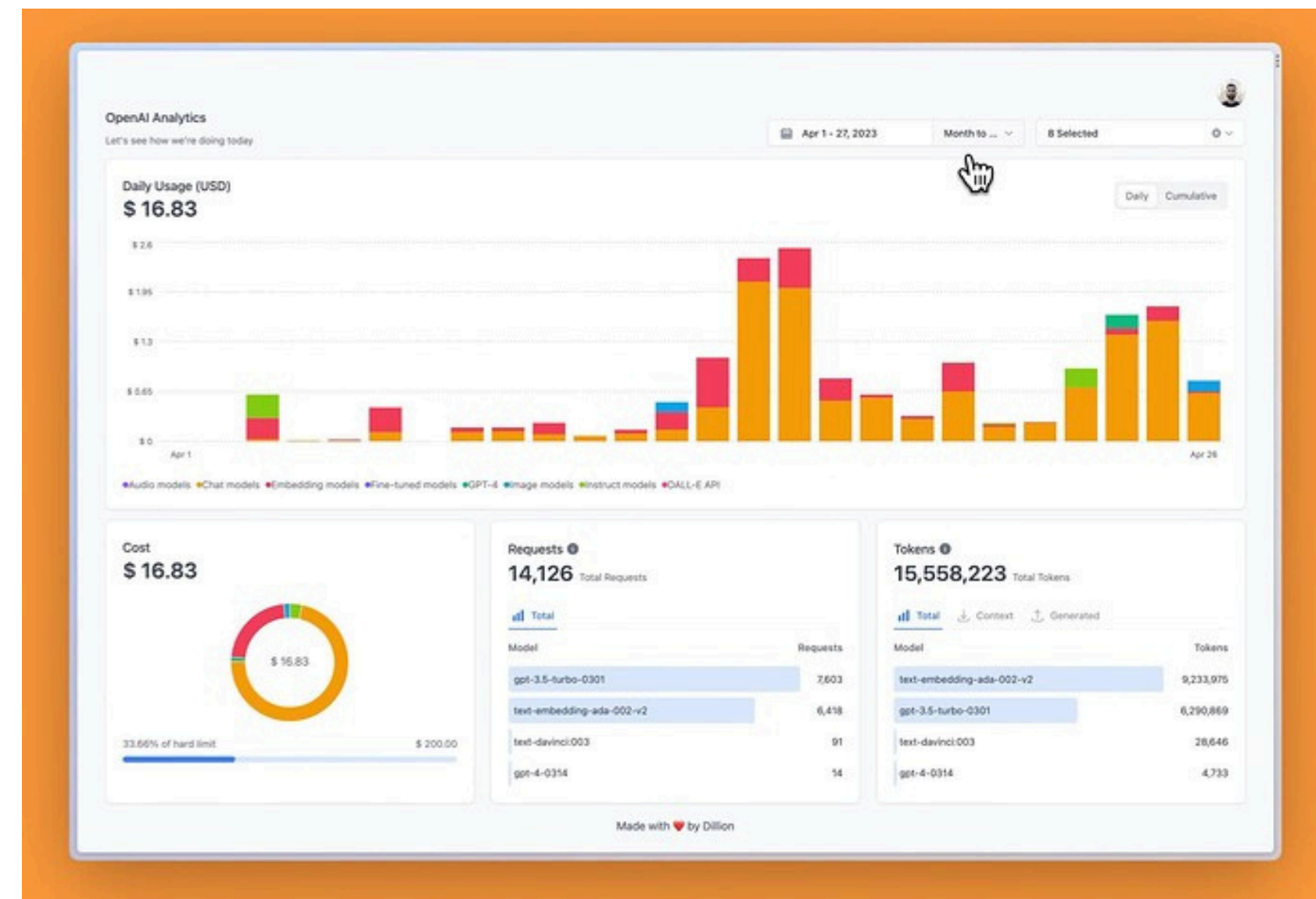
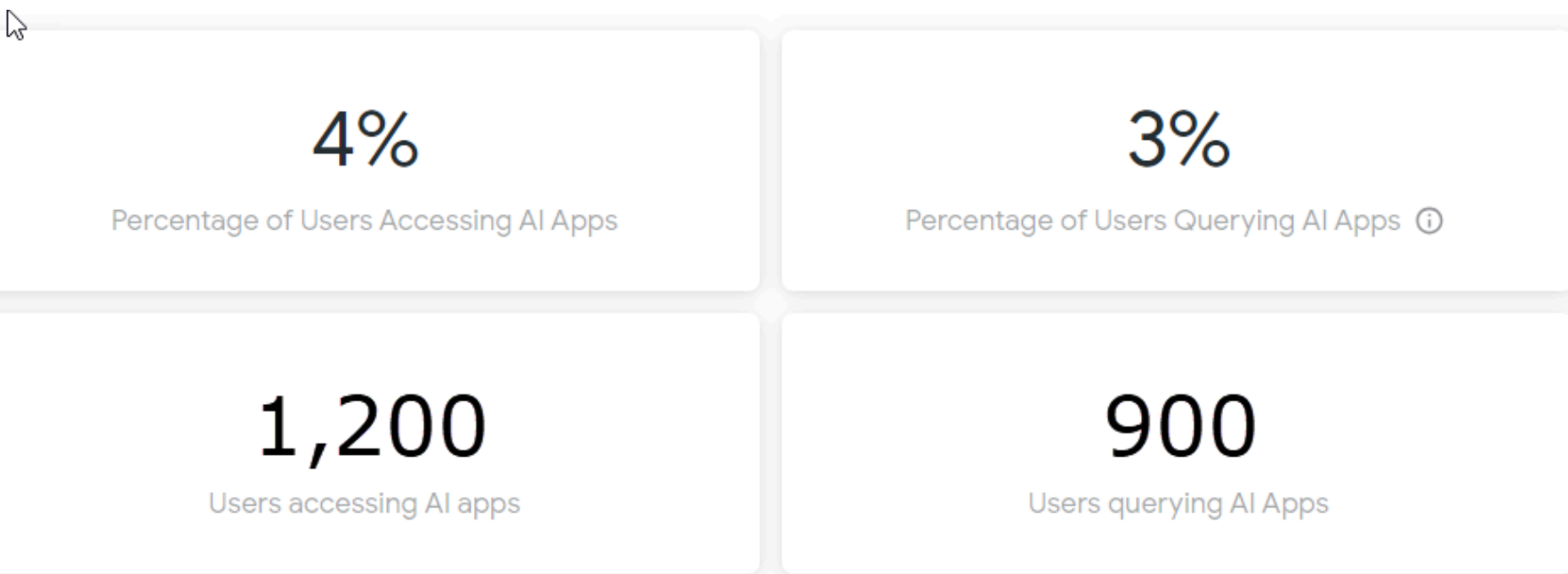
Mean: Add up values and divide by number of data points

Median: Middle value of data when ranked

Mode: Figure that appears most often in the data

Also, Percentages, Frequencies etc.

Descriptors and Charts - Visualizations of findings



How quantitative analysis supports personas, journey maps, and task flows

Supporting Personas

Quantitative data validates who your primary users really are.

Example:

Survey results showing 70% of users are students aged 18–25
→ Persona reflects this dominant group.

Metrics like frequency of feature use or time spent help flesh out persona behaviors.

How quantitative analysis supports personas, journey maps, and task flows

Supporting Journey Maps

Quantitative data highlights where bottlenecks occur.

Example:

Analytics show drop-offs at checkout (45%)

→ Journey map emphasizes frustration at the Payment stage.

Time metrics or Clickstream or heatmap data reveals which steps take longest.

How quantitative analysis supports personas, journey maps, and task flows

Supporting Task Flows / Task Analysis

Quantitative data measures efficiency and errors.

Example:

Task completion rate (e.g., only 60% succeed on first try)

→ Indicates usability issues.

Average time-on-task shows where workflows can be streamlined.

Reflecting on Quantitative Data

Strength	Example	Implication
Large Sample Sizes	Survey with 2,000+ respondents across demographics	Findings are more reliable and generalizable
Clear Patterns & Trends	Analytics show 80% of users drop off at checkout stage	Helps identify where to focus design improvements
Objective Evidence	Task completion times compared across two prototypes	Supports evidence-based design decisions

Reflecting on Quantitative Data

Limitation	Example	Implication
Missing Data	Survey has 200 responses but age	Cannot analyze age-based patterns;
Limited Time Range	App usage tracked only during exam	Findings may not generalize to normal
Data Accuracy Issues	Analytics inflated by bots / duplicate	Leads to false conclusions about
Not User-Validated	Checkout takes 5 mins : unclear if	Misinterpreting intent may cause

Always interpret charts with caution. Pair quantitative findings with qualitative insights to see the full picture. Quantitative data tells us what happened, but only users can tell us why it happened.

Reflecting on Qualitative Data

Strengths	Example	Implication
Rich, detailed insights	Interview reveals <i>why</i> users abandon	Provides context behind user behavior,
Captures emotions & motivations	Observation shows frustration facial	Helps design for emotional experience,
Uncovers unexpected findings	Open-ended survey response reveals users	Can inspire new features or product
Strong user empathy	Shadowing a user shows their real-life	Keeps design grounded in real contexts

Reflecting on Qualitative Data

Limitations	Example	UX Impact
Small sample sizes	Only 6–10 users interviewed	Findings may not be generalizable
Subjective interpretation	Researcher interprets a user’s	Risk of bias in design decisions
Time & resource intensive	Conducting 1-hour interviews with 20	Slower design cycles
Harder to quantify	User says “checkout is confusing,” but no	Need quantitative data to confirm scale

Summary : From Data to Defensible Design

The Analysis Stack

- Persona → Who
- Journey → Where problems happen
- Task → How users act
- Theme → What patterns explain behavior
- Quantitative → How much / how often

Each layer of the stack strengthens the next.

You do not need to do in strictly this order. It can be done in parallel as well.

Typically Persona would be done before Journey and Task Identification/

Summary : for CS3240 projects

What Strong Projects Do

- Trace every feature back to user research
- Show how persona shapes task structure
- Identify decision points & breakdowns
- Turn themes into design ideas
- Use quantitative data to validate priorities
- Justify trade-offs clearly

What Weak Projects Do

- Produce artifacts without linkage
- Write personas that don't influence tasks
- List task steps without analysis
- Cluster themes that never affect design
- Present numbers without interpretation

**In CS3240 projects, artifacts are not graded for completeness.
They are graded for the decisions they enable.**

Appendix

User Tasks

A few examples from student projects

These examples are derived from user research data.

While they provide useful insights, they are not flawless.

Critically analyze and consider potential improvements in view of the general guidelines (typically it should also be done in view of the user data context that is absent here with examples).

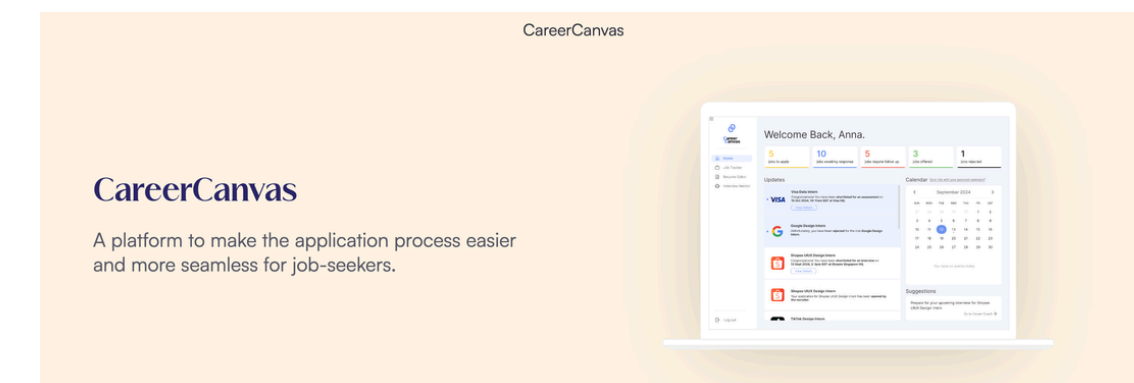
Example

1. Track application of all job applications across different job application sites to reduce cognitive load

The main function we propose is a **centralised platform** that allows job applicants to collate, track and manage their applications across all platforms with a **user-friendly and intuitive interface** that is easy to navigate.

We will focus on creating a **clean and organised dashboard** for tracking applications, setting reminders and accessing tips and insights.

<https://uxfol.io/p/96310685/02c3b2a2>

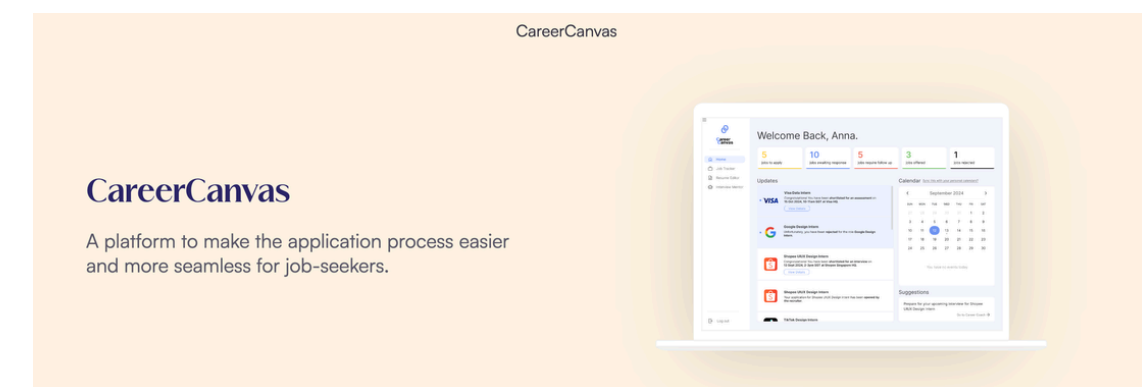


Example

2. Features for evaluating and improving resume writing based on a specific job posting

The second function that we propose is a **resume analysis assistant** which will be able to **identify potential gaps** between the job requirements and the user's resume. This tailored approach ensures that the user's application materials **showcase their strengths and skills optimally**, while also **pinpointing areas where the resume can be improved**.

<https://uxfol.io/p/96310685/02c3b2a2>



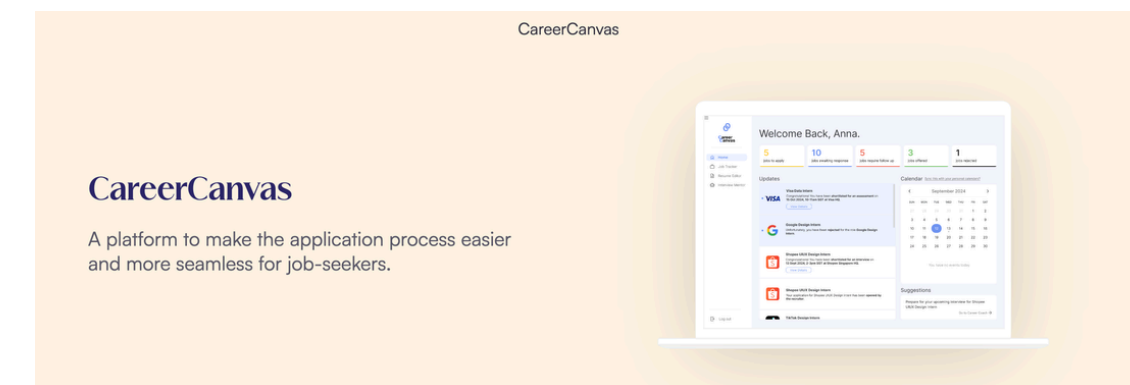
Example

3. Personalised job-seeking tips/advice to improve users' job search and application experience*

The last function function that we propose is an **AI career advisor**. It will provide common interview questions tailored to the specific company or industry and also offer **guidance** on how to **answer interview questions effectively**. Additionally, users will be able to use it to **stay informed about industry trends** and **adapt their job seeking strategies** for success.

** We included our original user task to show our process but this user task was changed along the way.*

<https://uxfol.io/p/96310685/02c3b2a2>



Example

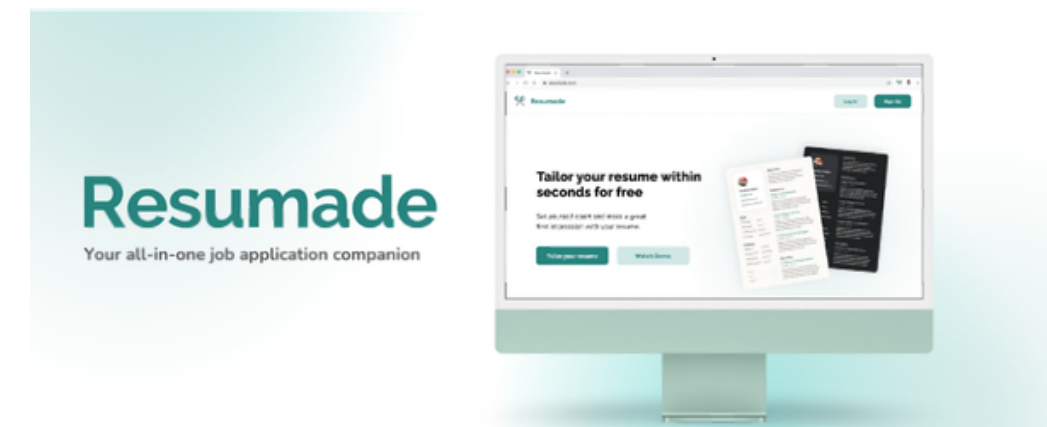
Task 1 - Job Description Analysis

Our app should make it easy for users to immediately extract key information from a job description and have keywords and important requirements highlighted to decide whether they are suitable for this job.

Frustration Targeted

Across the board, users have expressed that they spend a lot of time reading and analyzing the job description, to manually identify the exact requirements, job scope, and keywords, before making any relevant changes on their resume.

<https://uxfol.io/p/a4885f92/034c91f0>



Example

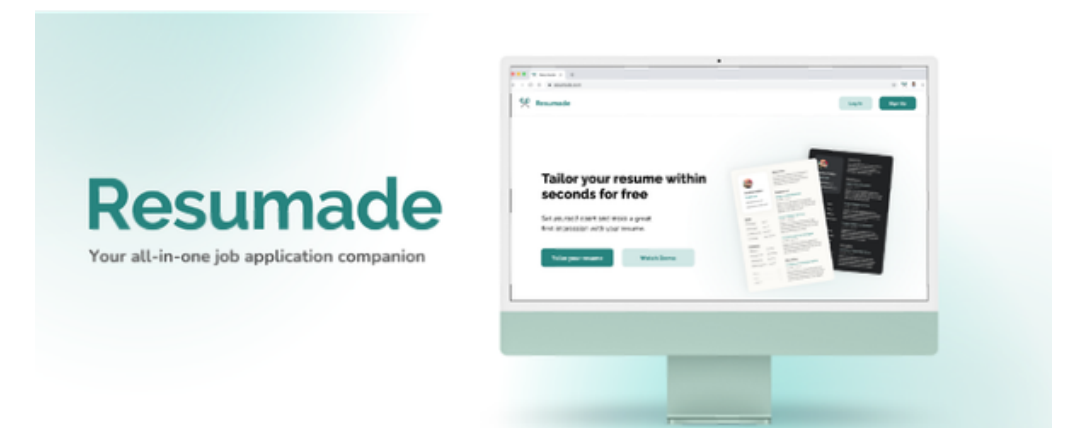
Task 2 - Resume Tailoring

With the assistance of our app, users should be able to receive suggestions of better ways to structure/order their resumes. They can choose appropriate action words that convey impact and demonstrate other positive and noteworthy qualities in their work and project experiences while tailoring their resume.

Frustration Targeted

Users conveyed to us that tailoring resumes is a time-consuming process that requires a lot of effort. From deciding how to phrase their bullet points to how to order the different sections of their resume, the entire process becomes tiresome rather quickly.

<https://uxfol.io/p/a4885f92/034c91f0>



Example

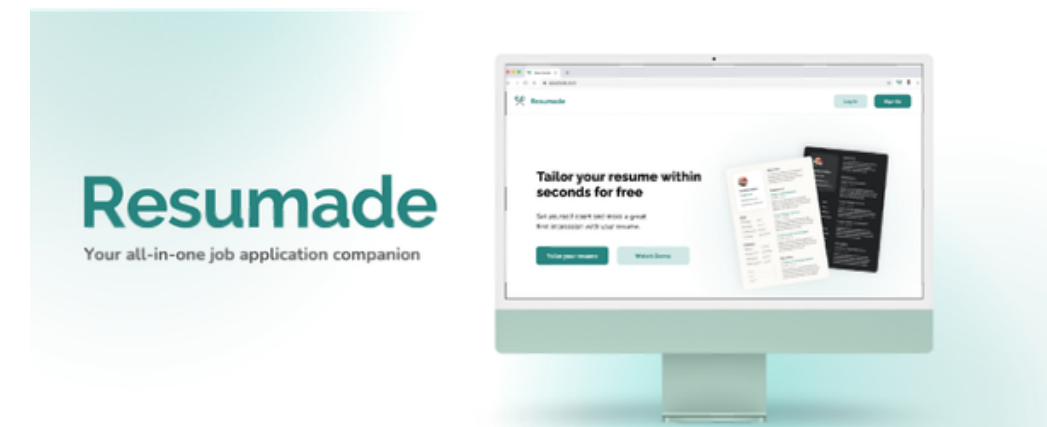
Task 3 - Application Tracker

Users would be able to keep track of their applications with an application tracking table containing internship/job description, weblink, resume, cover letter, etc.

Frustration Targeted

Users shared that having to keep track of a large number of applications can become messy very quickly, and they may even forget to add some applications to their tracking system.

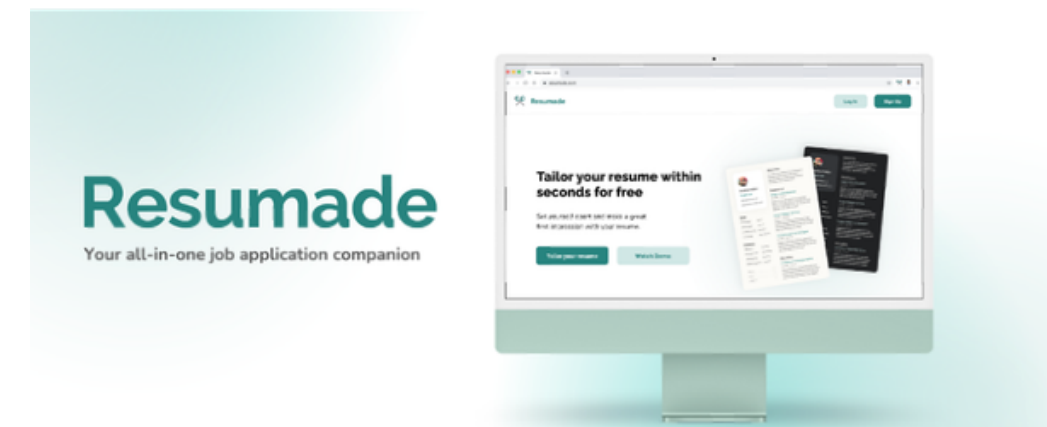
<https://uxfol.io/p/a4885f92/034c91f0>



Crafting Effective Task Titles

Task 1 - **Job Description Analysis**

Our app should make it easy for users to immediately extract key information from a job description and have keywords and important requirements highlighted to decide whether they are suitable for this job.



Crafting Effective Task Titles

- Titles should reflect an action or decision the user makes, rather than just a static description.
- The title should be written from the user's perspective, reflecting their experience, concern, or objective.
- Keep titles informative yet concise (usually under 10 words).
 - If the task is a problem, the title should reflect urgency or difficulty.
 - If the task is a solution, the title should emphasize ease or benefit.

Navigating a Tight Travel Schedule: Taré's Meal Dilemma

This title highlights both the logistical challenge of the journey and the specific problem Taré faces regarding food.

- Highlights both the travel constraint and the food challenge.

Seamless Travel: Pre-Ordering a Meal for a Tight Connection

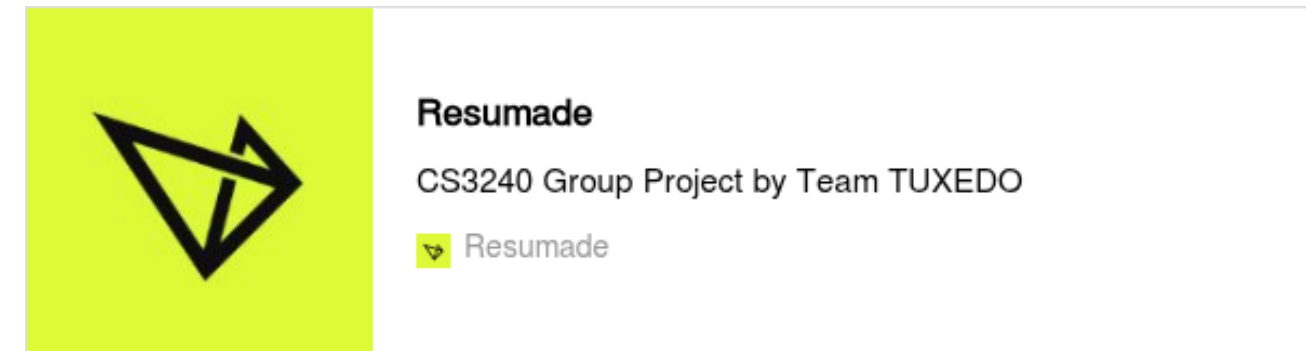
This title emphasizes both the solution (pre-ordering a meal) and the context (tight travel schedule), making it clear and engaging.

Focuses on the convenience of solving a specific travel challenge

Affinity Diagram - Examples from students' projects



<https://uxfol.io/p/96310685/02c3b2a2>



<https://uxfol.io/p/a4885f92/034c91f0>

*The examples here derived from user research data.
While they provide useful insights, they are not flawless representations.
Critically analyze and consider potential improvements in view of the general guidelines and of the user data context .*

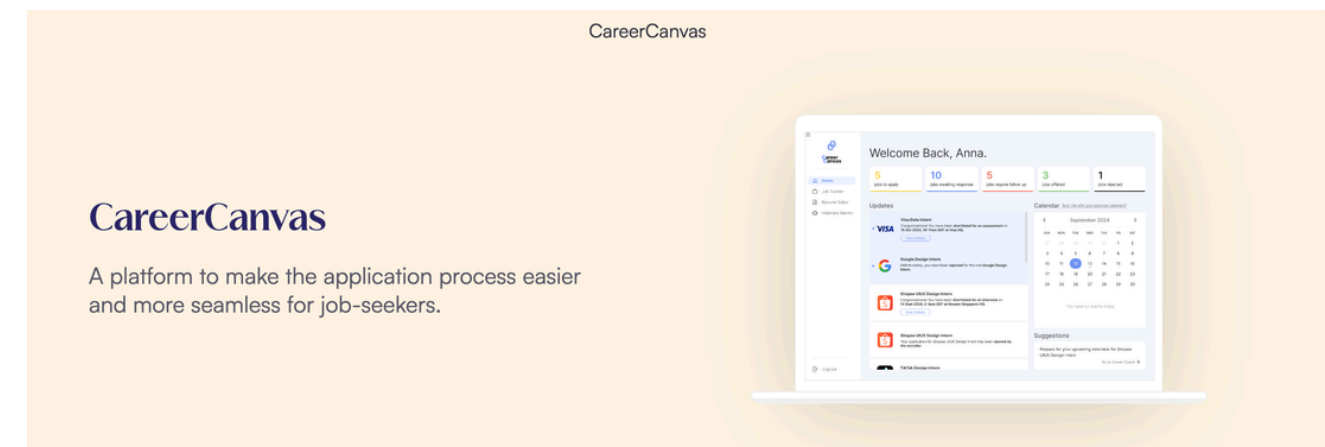
Example

....

Due to the sheer number of notes to be organised from our individual findings, **five initial high-level categories** namely “Job Search”, “Job Application”, “Job Tracking”, “Pre-Interview” and “Post-Interview” were defined in advance to facilitate the creation of the affinity diagram. The separation of the job-seeking process into these five categories created a structured framework for analysing and optimising each stage. It allows for a more detailed examination of user experiences, pain points, and needs at every step.

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<https://uxfol.io/p/96310685/02c3b2a2>



Affinity Diagram - reference



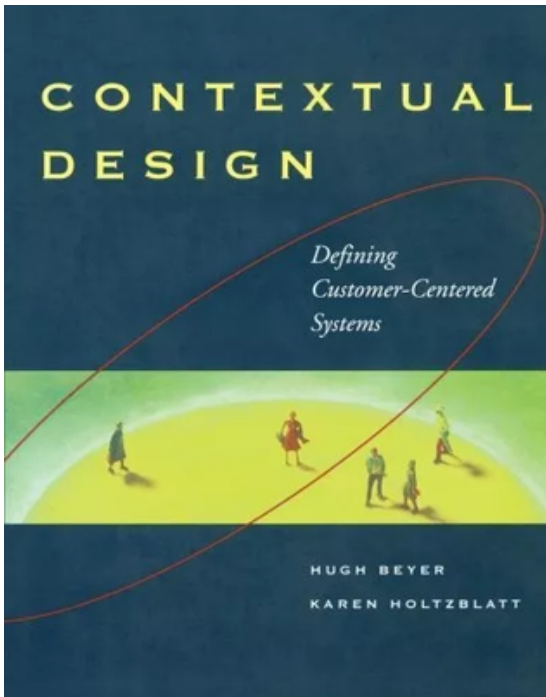
Anthropologist
Jiro Kawakita



Anthropologist
Karen Holtzblatt

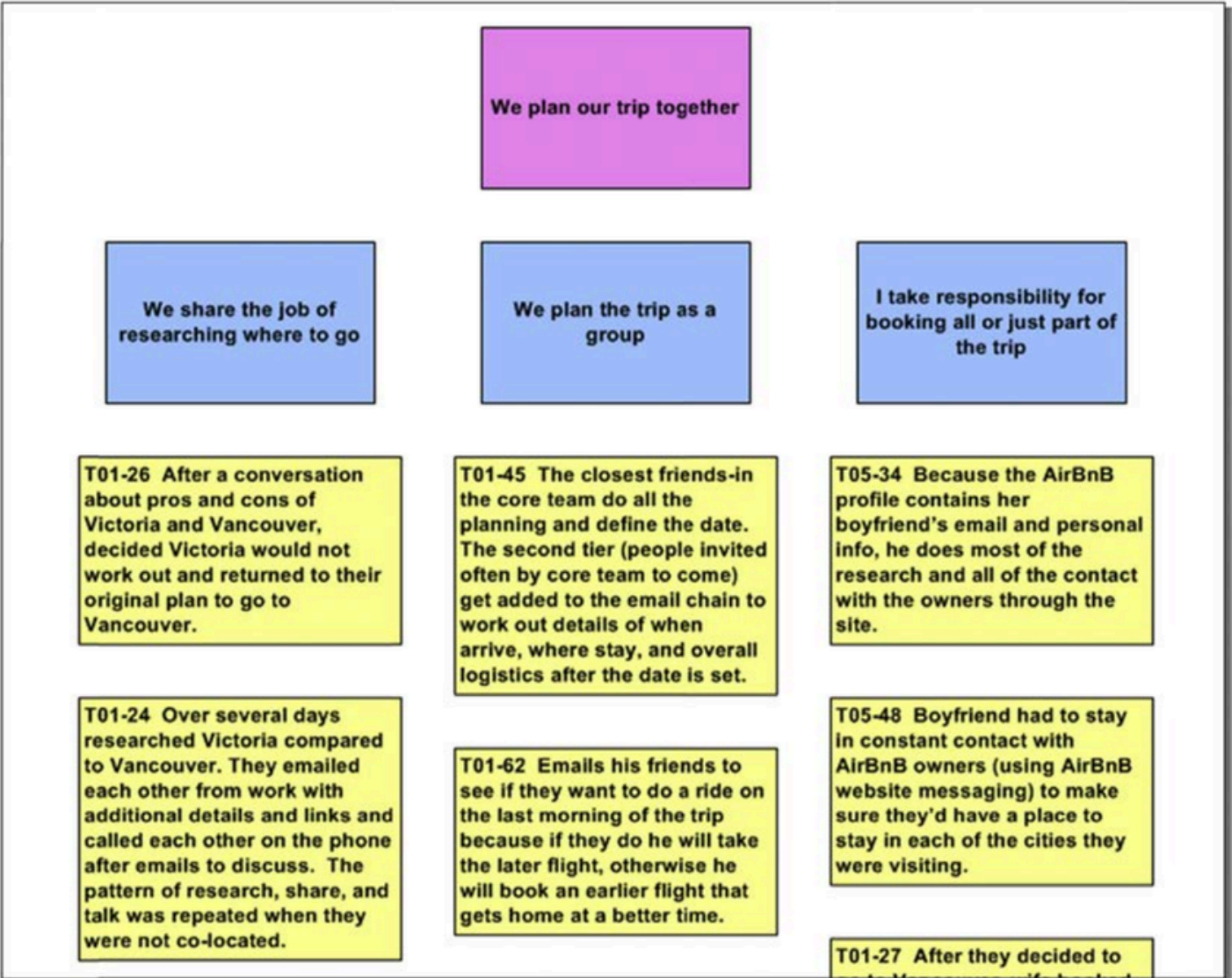


Psychologist
Hugh R. Beyer



6 - The Affinity Diagram

Karen Holtzblatt, Hugh Beyer



Tools for Data Analysis

‘Excel’ – quantitative analysis

SAS and SPSS – Statistical tools for Quantitative analysis

Nvivo – qualitative analysis tool.

It supports annotation and coding of data including PDfs, photos, Videos and Audio files

Dedoose – Cloud based, collaborative qualitative data management, excerpting/coding, and analysis tool